2.12 Explain why, when there is an electric current in a resistor, there is an energy transfer which heats the resistor

2.13 **Explain the energy transfer (in 2.12 above) as the result of collisions between electrons and the ions in the lattice**

2.14 Distinguish between the advantages and disadvantages of the heating effect of an electric current

2.15 Use the equation:

electrical power (watt, W) = current (ampere, A) × potential difference (volt, V)
$P = I \times V$

2.16 Use the equation:

energy transferred (joule, J) = current (ampere, A) × potential difference (volt, V) × time (second, s)
$E = I \times V \times t$

Topic 3

Motion and forces

3.1 Demonstrate an understanding of the following as vector quantities:

- a displacement
- b velocity
- c acceleration
- d force

3.2 Interpret distance/time graphs including determination of speed from the gradient

3.3 Recall that velocity is speed in a stated direction

3.4 Use the equation:

speed (m/s) = distance (m) / time (s)

3.5 Use the equation:

acceleration (metre per second squared, m/s^2) = change in velocity (metre per second, m/s) / time taken (second, s)

$$a = \frac{(v - u)}{t}$$

3.6 Interpret velocity/time graphs to:

- a compare acceleration from gradients qualitatively
- b calculate the acceleration from the gradient (for uniform acceleration only)
- c **determine the distance travelled using the area between the graph line and the time axis (for uniform acceleration only**

3.7 Draw and interpret a free-body force diagram

3.8 Demonstrate an understanding that when two bodies interact, the forces they exert on each other are equal in size and opposite in direction and that these are known as action and reaction forces

3.9 Calculate a resultant force using a range of forces (limited to the resultant of forces acting along a line) including resistive forces

3.10 Demonstrate an understanding that if the resultant force acting on a body is zero, it will remain at rest or continue to move at the same velocity

3.11 Demonstrate an understanding that if the resultant force acting on a body is not zero, it will accelerate in the direction of the resultant force

3.12 Demonstrate an understanding that a resultant force acting on an object produces an acceleration which depends on:

a the size of the resultant force

b the mass of the object

3.13 Use the equation:

force (newton, N) = mass (kilogram, kg) × acceleration (metre per second squared, m/s^2)
$F = m \times a$

3.14 Use the equation:

weight (newton, N) = mass (kilogram, kg) × gravitational field strength (newton per kilogram, N/kg)
$W = m \times g$

3.15 *Investigate the relationship between force, mass and acceleration*

3.16 Recall that in a vacuum all falling bodies accelerate at the same rate

3.17 Demonstrate an understanding that:

a when an object falls through an atmosphere air resistance increases with increasing speed

b air resistance increases until it is equal in size to the weight of the falling object

c when the two forces are balanced, acceleration is zero and terminal velocity is reached

Topic 4

Momentum, energy, work and power

4.1 Recall that the stopping distance of a vehicle is made up of the sum of the thinking distance and the braking distance

4.2 Demonstrate an understanding of the factors affecting the stopping distance of a vehicle, including:

a the mass of the vehicle

b the speed of the vehicle

c the driver's reaction time

d the state of the vehicle's brakes

e the state of the road

f the amount of friction between the tyre and the road surface

4.3 *Investigate the forces required to slide blocks along different surfaces, with differing amounts of friction*

4.4 Use the equation:

momentum (kilogram metre per second, kg m/s) = mass (kilogram, kg) × velocity (metre per second, m/s)

to calculate the momentum of a moving object

4.5 Demonstrate an understanding of momentum as a vector quantity

4.6 Demonstrate an understanding of the idea of linear momentum conservation

4.7 Demonstrate an understanding of the idea of rate of change of momentum to explain protective features including bubble wraps, seat belts, crumple zones and air bags

4.8 *Investigate how crumple zones can be used to reduce the forces in collisions*

4.9 **Use the equation:**

force (newton, N) = change in momentum (kilogram metre per second, kg m/s) / time (second, s)

$$\boldsymbol{F = \frac{(mv - mu)}{t}}$$

to calculate the change in momentum of a system, as in 4.6

4.10 Use the equation:

work done (joule, J) = force (newton, N) × distance moved in the direction of the force (metre, m)

$E = F \times d$

4.11 Demonstrate an understanding that energy transferred (joule, J) is equal to work done (joule, J)

4.12 Recall that power is the rate of doing work and is measured in watts, W

4.13 Use the equation:

power (watt, W) = work done (joule, J) / time taken (second, s)

$P = \frac{E}{t}$

4.14 Recall that one watt is equal to one joule per second, J/s

4.15 Use the equation:

gravitational potential energy (joule, J) = mass (kilogram, kg) × gravitational field strength (newton per kilogram, N/kg) × vertical height (metre, m)

GPE $= m \times g \times h$

4.16 Use the equation:

kinetic energy (joule, J) = $\frac{1}{2}$ × mass (kilogram, kg) × velocity^2 ((metre/second)^2 (m/s)^2)

KE $= \frac{1}{2} \times m \times v^2$